[2026] ML & PR Projects – Milestone 2

The objective of the projects is to prepare you to apply different machine learning algorithms to real-world tasks. This will help you to increase your knowledge about the workflow of the machine learning tasks. You will learn how to apply pre-processing, feature engineering, regression, and classification methods.

- **Delivering Milestone 2: Practical exam.**

- You must deliver a detailed report for milestone 2 contains all your work in this phase. Combine both reports and deliver a complete report for the project (Hardcopy).

- Each team should work on their project's updated dataset for milestone 2.

- **In the practical exam**:

    - We will give you two unseen test sets, **one for regression and one for classification.**

    - Make sure you **save your trained model** and create a test script that takes the new csv file, **loads the saved models,** and outputs predictions. This is to allow us to test your model without re-training.

      Hint 1: You can use libraries such as 'pickle' to save and load your models.
      Hint 2: Any technique (preprocessing or otherwise) that you need to 'fit' or 'learn' during training means you need to save it and reload it for the test to work correctly.

    - You should be able to handle missing values for features in a test sample. (You can't drop an entire test sample row).

    - You must Show the MSE and R2 score of the regression models and the classification accuracy of each classifier on the test set.

- Each team member will be graded individually according to their response to the oral questions related to their project.

- In the second milestone, you will apply the following: -

  Classification:

  - Split your dataset into 80% training and 20% testing.

  - Train at least 3 different models to classify each sample into distinct classes.

  - Choose at least two hyperparameters to vary. Study **at least three different choices** for each hyperparameter. When varying one hyperparameter, all the other hyperparameters should be fixed.

Milestone 2:

- Classification and Hyperparameter tuning.

Milestone 2 Report Must Include:

- Summarize the **classification accuracy**, **total training time**, and **total test time** using three bar graphs.

- Note that your **Feature Selection** process may differ in this phase (classification) than the previous (regression), If so, explain your feature selection process and how it was proved or disproved.

- Explain in details how **hyperparameter tuning** affected your models' performance.

- Finally, write a **conclusion** about this phase of the project and what intuition you had about your problem and how it was proved/disproved.

Project(1)-SC-1&-AI-Rob-1: Online Games Popularity Prediction
An **updated dataset** will be provided for each project in the second milestone.

Updated Dataset Snapshots:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
QueryID	ResponseI	QueryNam	ResponseI	ReleaseDa	RequiredA	DemoCou	Developer	DLCCount	Metacritic	MovieCou	PackageC	GamePopularity	PublisherC	Screensho
10	10	Counter-S	Counter-S	Nov 1 200(	0	0	1	0	88	0	1	High	1	13
20	20	Team Forti	Team Forti	Apr 1 1999	0	0	1	0	0	0	1	High	1	5
30	30	Day of Def	Day of Def	May 1 200;	0	0	1	0	79	0	1	High	1	5
40	40	Deathmat	Deathmat	Jun 1 2001	0	0	1	0	0	0	1	Medium	1	4
50	50	Half-Life: (	Half-Life: (	Nov 1 1999	0	0	1	0	0	0	1	High	1	5
60	60	Ricochet	Ricochet	Nov 1 200(	0	0	1	0	0	0	1	High	1	4
70	70	Half-Life	Half-Life	Nov 8 1998	0	0	1	1	96	0	1	High	1	11
80	80	Counter-S	Counter-S	Mar 1 2004	0	0	1	0	65	0	1	High	1	8
130	130	Half-Life: E	Half-Life: E	Jun 1 2001	0	0	1	0	71	0	1	High	1	5
220	220	Half-Life 2	Half-Life 2	Nov 16 200	0	1	1	1	96	2	2	High	1	9
240	240	Counter-S	Counter-S	Nov 1 2004	0	0	1	0	88	0	1	High	1	5

Updated Dataset Description:

- The **"RecommenderCount"** column used in the previous milestone as the actual output has been removed.
- A New column is added **"GamePopularity"**. The categories can be: {Low, Medium or High}.

Milestone 2 Classification task:

Classify into one of three categories: {Low, Medium or High}.
based on the provided features in **the updated dataset.**

Project(2)-SC-2: E-Learning Student Performance Prediction

An **updated dataset** will be provided for each project in the second milestone.

Updated Dataset Snapshot:

id_assessı	id_student	date_subn	is_banked	assessment
24295	335914	21	0	Very Good
34910	648903	235	0	Good
25353	103800	207	0	Fail
25337	537926	116	0	Fail
37417	575559	171	0	Very Good
15003	599639	56	0	Excellent
25365	104772	110	0	Very Good
30719	677947	32	0	Very Good
34910	693628	236	0	Good
24291	552498	31	0	Excellent
1752	229361	18	0	Very Good
15022	695298	110	0	Good

Updated Dataset Description:

- The **"score"** column used in the previous milestone as the actual output has been removed.

- A New column is added **"assessmentClass"**. The categories can be: {fail, good, very good or excellent}.

Milestone 2 Classification task:

Classify into one of four categories: fail, good, very good or excellent based on the provided features in **the updated dataset.**

Project(3)-CS-1: Software Developers Job Satisfaction Prediction

An **updated dataset** will be provided for each project in the second milestone.

Updated Dataset Snapshot:

AIAgentKn	AIAgentOr	AIAgentOr	AIAgentOb	AIAgentOb	AIAgentExt	AIAgentExt	AIHuman	AIOpen	Converted	JobSatLevel
is a barrier	Vertex AI				ChatGPT		When I dor	Troubleshc	61256	High
sting tools and workflows can be difficult.;My company's IT and/or InfoSec tea							When I dor	All skills. A	104413	High
					ChatGPT;Claude Cod		When I dor	Understan	53061	Average
sting tools and workflows can be difficult.					ChatGPT;Claude Cod		When I donâ€™t trust .		36197	Low
'or InfoSec teams have strict rules that do not allow me t					GitHub Copilot		When I hav	designing t	120000	Average
rt to learn how to use AI agents efi			Sentry		ChatGPT;Claude Cod		When I donâ€™t trust .		6240	Low
with my existing tools and workflows can be difficult.;My company's IT and/or I							When I wai	Innovation	72000	Low
									70000	Average
'or InfoSec teams have strict rules that do not allow me to use AI agent tools oı							When Iâ€™m stuck an		5400	Low

Updated Dataset Description:

- The **“JobSat”** column used in the previous milestone as the actual output has been removed.
- A New column is added **“JobSatLevel”**. The categories can be: {Low, Average or High}.

Milestone 2 Classification task:

Classify into one of three categories: {Low, Average or High}based on the provided features in **the updated dataset.**

Project(4)-CS-2: Movie Popularity Prediction

An **updated dataset** will be provided for each project in the second milestone.

Updated Dataset Snapshot:

backdrop_	budget	homepage	imdb_id	original_la	original_tit	overview	popularityLevel
/1syW9SN	1.2E+08	https://ww	tt9362930	en	Blue Beetl	Recent col	High
/xFYpUmB	60000000	https://ww	tt4495098	en	Gran Turisı	The ultima	High
/53z2fXEK	38500000	https://ww	tt1016097	en	The Nun II	In 1956 Fra	High
/5mzr6JZb	1.29E+08	https://ww	tt9224104	en	Meg 2: The	An explora	High
/iiXliCeykk	20000000	https://iiil.i	tt6906292	en	Retributior	When a my	High
/ilvQnZyzg	4500000	https://um	tt1063852	en	Talk to Me	When a grc	High
/4XM8DUT	3.4E+08	https://fas	tt5433140	en	Fast X	Over many	High
/eMPxmNv	15000000	https://ww	tt7599146	en	Sound of F	The story o	High
/ctMserH8	1.45E+08	https://ww	tt1517268	en	Barbie	Barbie and	High

Updated Dataset Description:

- The **"popularity"** column used in the previous milestone as the actual output has been removed.
- A New column is added **"popularityLevel"**. The categories can be:{Very Low, Low, Medium or High}.

Milestone 2 Classification task:

Classify into one of four categories: {Very Low, Low, Medium or High} based on the provided features in **the updated dataset.**

Project(5)-AI-Rob-2: Vitamin Deficiency Prediction

An **updated dataset** will be provided for each project in the second milestone.

Updated Dataset Snapshot:

sun_expos	income_le	latitude_re	vitamin_a_	vitamin_c_	vitamin_d_	vitamin_e_	vitamin_b1	folate_per	calcium_p	iron_perce	symptoms	symptoms	disease_diagnosis
High	High	Mid	119.1	147.3	152.88	97.5	102.5	188.9	108.3	97.4	0	None	Healthy
Low	Low	Low	85.7	57.5	32.76	82.7	62.6	51	42.6	102.5	1	bone_pain	Rickets_Osteomalac
Low	High	High	48.3	152.1	94.99	169.3	136.2	116.6	136.3	86.4	2	dry_skin;ni	Healthy
High	Low	Low	75.8	51	51.48	85.7	31.8	66.5	76.5	60.8	2	numbness	Anemia
Moderate	High	Low	93.3	111.5	62.9	155.6	72.6	124.9	69.4	71.9	0	None	Healthy
Moderate	Low	Mid	47.9	75.3	65.5	72.2	21.2	101	70	79	3	numbness	Anemia
Low	Low	High	62.2	106.7	25.27	53.7	27	77.4	67.1	54.7	2	bone_pain	Rickets_Osteomalac
Low	High	Mid	129.6	138	42.56	70.8	50	162.1	151.2	120.4	2	bone_pain	Rickets_Osteomalac
High	High	Low	192.5	123.5	166.01	84.7	125.9	98.1	198.9	40.3	1	fatigue	Anemia
Moderate	Middle	Mid	111.2	108.3	111.8	67.8	88.7	104.7	89.7	77	0	None	Healthy

Updated Dataset Description:

- The **"vitamin_deficiency"** column used in the previous milestone as the actual output has been removed.

- A New column is added **"disease_diagnosis"**. The categories can be: { scurvy, anemia, healthy, Rickets_Osteomalacia, Night blindness}.

Milestone 2 Classification task:

Classify into one of five categories: { scurvy, anemia, healthy, Rickets_Osteomalacia, Night blindness} based on the provided features in **the updated dataset.**